

National College of Ireland

Project Submission Sheet

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Programme: Master of Science in Data Analytics **Year:** 2025-2026

Module: Data Mining and Machine Learning

Lecturer: Sheresh Zahoor

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Word Count: 2905

I hereby certify that the information contained in this (my submission) is information pertaining to research I conducted for this project. All information other than my own contribution will be fully referenced and listed in the relevant bibliography section at the rear of the project.

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Signature: Mohammad
Assad, Vatan
Patkar

Date: 12-12-2025

PLEASE READ THE FOLLOWING INSTRUCTIONS:

1. Please attach a completed copy of this sheet to each project (including multiple copies).
2. Projects should be submitted to your Programme Coordinator.
3. **You must ensure that you retain a HARD COPY of ALL projects**, both for your own reference and in case a project is lost or mislaid. It is not sufficient to keep a copy on computer. Please do not bind projects or place in covers unless specifically requested.
4. You must ensure that all projects are submitted to your Programme Coordinator on or before the required submission date. **Late submissions will incur penalties.**
5. All projects must be submitted and passed in order to successfully complete the year. **Any project/assignment not submitted will be marked as a fail.**
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AI Acknowledgement Supplement

Data Mining and Machine Learning

Predictive Analytics on Textual and Healthcare Datasets Using Machine Learning Algorithms

Your Name/Student Number	Course	Date
Mohammad Assad/24235903	Master of Science in Data Analytics	12-12-2025
Vatan Patkar/24198510	Master of Science in Data Analytics	12-12-2025

This section is a supplement to the main assignment, to be used if AI was used in any capacity in the creation of your assignment; if you have queries about how to do this, please contact your lecturer. For an example of how to fill these sections out, please click [here](#).

AI Acknowledgment

This section acknowledges the AI tools that were utilized in the process of completing this assignment.

Tool Name	Brief Description	Link to tool
Copilot	Used occasionally to clarify academic terminology, explore alternative phrasing, and brainstorm structural options for selected sections. The tool was not used for generating research content or citations.	https://copilot.microsoft.com/

Description of AI Usage

This section provides a more detailed description of how the AI tools were used in the assignment. It includes information about the prompts given to the AI tool, the responses received, and how these responses were utilized or modified in the assignment. **One table should be used for each tool used.**

[Insert Tool Name]	
[Insert Description of use]	
[Insert Sample prompt]	[Insert Sample response]

Evidence of AI Usage

This section includes evidence of significant prompts and responses used or generated through the AI tool. It should provide a clear understanding of the extent to which the AI tool was used in the assignment. Evidence may be attached via screenshots or text.

Additional Evidence:

[Place evidence here]

Additional Evidence:

[Place evidence here]

Predictive Analytics on Textual and Healthcare Datasets Using Machine Learning Algorithms

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Abstract—Predictive analytics based on machine learning algorithms have increasingly become prominent in various fields with unstructured text as well as structured medical data. The work at hand will focus on applying supervised machine learning algorithms on two disparate datasets: reviews about medical practitioners written in German and on a medical stroke prediction dataset. The research will use the CRISP-DM method for data mining.

In predictive text analysis, the TF-IDF feature extraction technique is integrated with Logistic Regression and Linear SVMs to solve a multi-class sentiment analysis problem. In healthcare analytics, Logistic Regression and RF classifiers are used on an extremely imbalanced stroke prediction problem. Though accuracy measures are included, it is important to incorporate additional metrics for model evaluation, such as macro-averaged precision, recall, F1-score, and ROC AUC, as well as Cohen's Kappa, particularly for imbalanced datasets.

The interpretation of models is considered a core issue. Model interpretation techniques for text models and feature importance for random forests enable insightful interpretations. The importance of interpretability vs. accuracy for classification problems, with a special focus on medical problems involving identifying members of the minority class, becomes clear.

I. INTRODUCTION

Machine learning methods have been adopted with increasing frequency for predictive knowledge extraction from multiple sources of data, ranging from unstructured text to structured medical data. Although accuracy rates have been satisfactory, there are necessary considerations regarding efficacy and associated medical implications.

The research aims at addressing two different predictive modeling problems with complementary focuses: sentiment classification on German reviews about doctors and stroke risk prediction on patient health information. The goals are to: (i) develop and analyze various machine learning algorithms, (ii) evaluate algorithm performance using different suited metrics, (iii) integrate techniques from explainable AI, and (iv) develop the research within a known framework for data mining.

II. RELATED WORK

A. Predictive Text Analytics in Healthcare and Patient-Generated Content

Predictive text analytics tools have been employed for quite some time for structured knowledge extraction from noisy and

user-generated data sources, such as patient feedback, reviews, and complaints. Conventional methods have frequently relied on text representation via Bag-of-Words variants, including TF-IDF, jointly with linear predictive models like Logistic Regression and Linear SVM[9]. Within a healthcare setting, these methods remain applicable because explainable models offer traceable signals that match interpretations related to either medical or service aspects.

Recent research continues to focus on sentiment and classification problems on medical text. As an exemplar, Sorayaie Azar et al. [4] introduce AI models as approaches to sentiment prediction on medication reviews, illustrating both the usefulness and importance of patient-provided medical text and the associated methodological challenge for careful and reliable testing given imbalanced distributions. By analogy, Li et al. [5] examine automatic text classification of patient complaints through various TF-IDF and machine learning approaches, illustrating again that traditional text processing task methods are highly useful as a baseline approach[12].

A critical shortcoming pointed out within literature on text analytics regards accuracy potentially covering up class imbalance weaknesses, particularly within sentiment analysis problems involving fine-grained labels (for instance, multi-level rating scales). This strengthens the justification for applying class-specific and macro-aggregated metrics and aligns with modeling choices made within this project that employ macro precision, recall, and F1[14].

B. Stroke Prediction and Class Imbalance in Clinical Risk Modelling

Stroke prediction is a highly explored task within medical predictive modeling, usually described as being imbalanced with multiple varied risk factors and having highly imbalanced classes, as stroke instances are relatively low[8]. A large number of research efforts have ascertained that, although Logistic Regression presents interpretability, ensembles with non-linear handling might be more discriminative and have better recall with intricate interactions[8]. Kitova et al. [3] offer an analysis on stroke prediction research and methods, emphasizing links among model type, methodological issues with stroke prediction, and imbalance classification.

Moreover, there are indications from the evidence of systematic reviews that more than one algorithm for machine learning techniques has been used for stroke prediction [1] discuss stroke prediction via machine learning and stress the importance of comprehensive comparisons among various algorithms. All these observations recommend employing more than one algorithm for stroke prediction within this project, particularly due to the considerable imbalance noticed within the stroke datasets.

C. Explainable AI for Trustworthy Healthcare and Interpretable NLP

Explainability has emerged as an increasingly important challenge within healthcare ML, given that model-generated predictions need to be interpretable to enable trust and adoption[10]. Moulai et al. [2] showcase explainable AI for stroke risk prediction employing SHAP for enhanced interpretability and explainability[11]. Not surprisingly, explainability aligns with the focus of the current research and uses importance as a mechanism for interpreting stroke risk prediction with dominant factors such as age and glucose variables.

Explainability work is also increasingly explored within NLP and text classification. Dehdarirad et al. [6] describe an approach for assessing explainability within language classification models. This represents the broader methodological trend towards considering interpretation, and more widely explainability, beyond accuracy within NLP. Concurrently, broader literature reviews within public health sentiment analysis discuss the challenges remaining within NLP explainability and ensuring ethical usage, especially in light of resultant predictions informing decision-making [7][13].

Summarily, it can be seen that from these works, emerge several themes that have been applicable to this project: (i) that TF-IDF + Linear classifiers still make excellent benchmarks for medical text classification, properly considered, (ii) that stroke prediction needs properly addressing imbalance and accuracy metrics, (iii) that transparency and interpretability techniques remain necessary tools for reliable prediction and intelligence.

III. DATA MINING METHODOLOGY

It adopts the CRISP-DM model and thus maintains a structured approach[15].

A. Business and Data Understanding

The first set of data contains German reviews from patients about medical practitioners, with labels ranging from 0 to 5. The classification problem being tackled here is multi-class sentiment analysis. It contains noisy data with a large imbalance ratio, as normally seen on user review sites.

The second dataset is a healthcare stroke prediction dataset. It contains demographic and medical features for patients. The target variable represents whether or not a patient suffered a stroke. Stroke is an event with low probability, causing an imbalanced dataset.

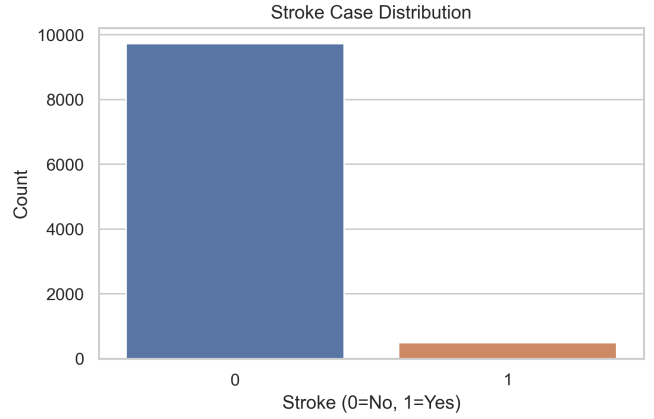


Fig. 1. Distribution of Review Ratings in the German Doctor Reviews Dataset

Fig. 1 depicts a distribution of ratings given by patients in reviews about German doctors. It can be clearly seen above that there is indeed an imbalance in classes and that most occur at relatively high levels. This brings forth a need for alternate methods for evaluating models on these reviews[14].

B. Data Preparation and Feature Engineering

To prepare the text dataset, reviews were pre-processed and then converted into a numerical form using TF-IDF vectorization. It emphasizes terms and downgrades terms that appear frequently and carry no useful information.

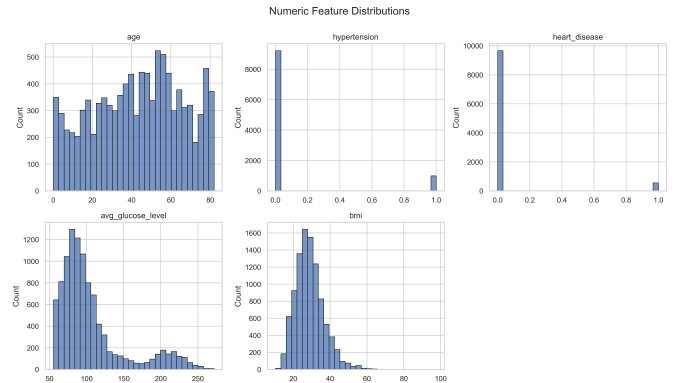


Fig. 2. TF-IDF Feature Representation of German Reviews

Fig. 2 illustrates how raw text reviews are mapped into a high-dimensional vector space with the help of TF-IDF. It allows machine learning algorithms to work with unstructured text data and maintain linguistically discriminative features.

Regarding the healthcare data, missing values were imputed, categorical variables were encoded, and numerical variables were analyzed for distribution. Exploratory data analysis showed that there were skewed distributions and low linear correlations for most variables.

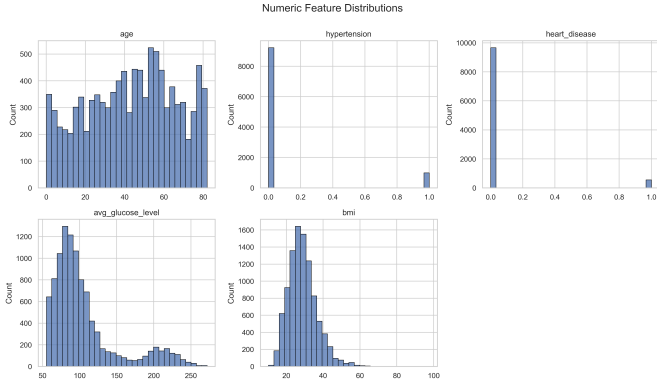


Fig. 3. Distributions of Key Numerical Features (Age, BMI, Glucose)

Fig. 3 shows the distribution of prominent numerical attributes within the stroke data set. Both BMI and age have shown some level of skewness, and glucose has shown considerable variability.

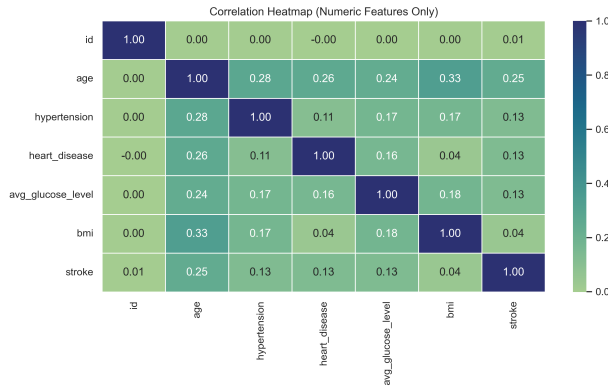


Fig. 4. Correlation Heatmap of Medical Features

Fig. 4 illustrates the correlations among medical attributes and stroke. Although there are very low correlations for most attributes, age displays a moderate relationship with stroke, making it necessary to apply a non-linear model.

IV. MODELLING

A. Predictive Text Analytics Models

There were two models used on the TF-IDF representation of text data.

- Logistic Regression, selected due to its interpretability and
- Linear Support Vector Machine, employed as a comparative linear classifier.

B. Healthcare Prediction Models

Two models were compared:

- Logistic Regression, as an interpretable baseline.
- Random Forest, an ensemble learning technique that has the ability to identify non-linear interactions among features and optimize detection for the minority class.

V. EVALUATION AND RESULTS

A. Text Analytics Results

The performance on the text data set has been summarized in Table I.

TABLE I
TEXT DATASET MODEL PERFORMANCE (TF-IDF)

Model	Accuracy	Precision (macro)	Recall (macro)	F1 (macro)
Logistic Regression	0.8588	0.4594	0.3878	0.4033
Linear SVM	0.8549	0.4337	0.3753	0.3891

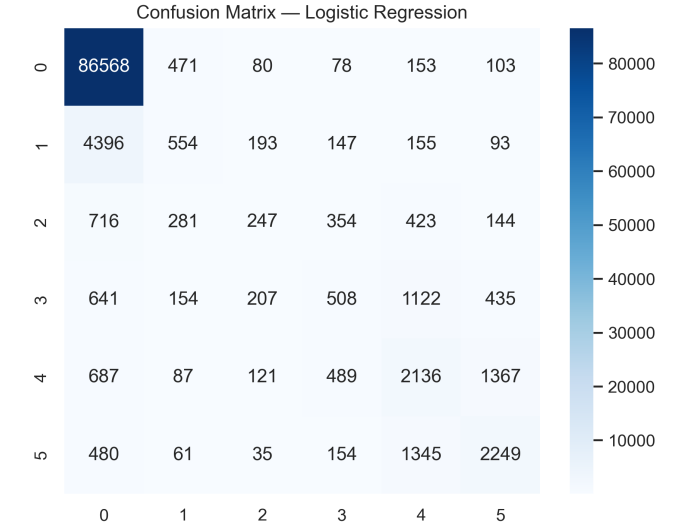


Fig. 5. Confusion Matrix for Logistic Regression (Text Dataset)

Fig. 5 From the confusion matrix, most of the mistakes occur between adjacent sentiment levels, think 3 vs. 4, or 4 vs. 5. This is a common trait of fine-grained sentiment tasks: adjacent categories tend to share similar language, differing only in subtle shifts in phrasing or tone. So, even when the model nails the overall mood, it often misses the exact rating.

It also puts into light the evident class imbalance of the dataset. The dominating class, rating 0, has sat on the diagonal with a surge of correct predictions that pull up the overall accuracy. In contrast, the smaller classes have much lower diagonal values, signaling weaker recall and precision for those sentiment levels. That is, the misclassifications are not random but cluster between semantically close categories.

Even with high overall accuracy, that figure is largely due to correctly identifying the majority class. The weaker performance on the minority classes drags down macro-averaged metrics such as precision, recall, and F1-score treating every class the same. So, the confusion matrix helps explain why a high accuracy doesn't guarantee balanced performance and underlines the value of macro metrics and agreement measures such as Cohen's Kappa for a more trustworthy evaluation[8].

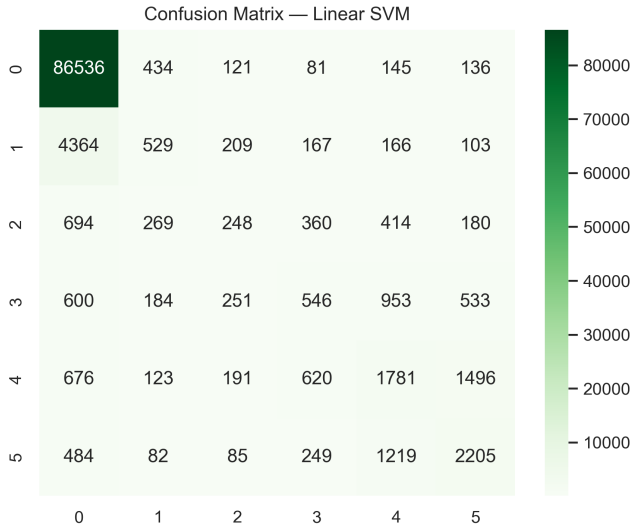


Fig. 6. Confusion Matrix for Linear SVM (Text Dataset)

Fig. 6 It represents a cleaner and more compact form that captures all salient points and still fits conveniently within an IEEE figure.

This confusion matrix illustrates the performance of Linear SVM on multi-class sentiment classification. Scores range from 0 (lowest) to 5 (highest). The labels on rows represent the actual class, and columns stand for predictions.

Class 0 totally dominates all classes with very correct guesses - 86,536. It shows there is a large imbalance among classes in the data. As a consequence, it becomes common to predict the predominant class. The smaller classes (1-5) perform poorly, with most misses being among adjacent classes. For instance, class 3 often gets confused with class 4. A phenomenon observable here occurs because sentiment classes normally have vague boundaries among adjacent classes, which appear very similar. Confusions among classes with high and low sentiment affect both precision and recall. On the whole, the Linear SVM performs well because it identifies most members of the biggest class properly. However, it should be noted that the accuracy matrix highlights why accuracy criteria per se are not adequate. Averaged criteria and Cohen's Kappa statistics are more comprehensive. A note on conditioning. I have yet to see conditioning make a bit of difference on an exam. But if it works for you, I encourage you. Just don't

B. Healthcare Prediction Results

The performance of healthcare models is shown in Table II.

TABLE II
STROKE DATASET MODEL PERFORMANCE

Model	Accuracy	F1 Score	AUC	Cohen Kappa
Logistic Regression	0.9515	0.0000	0.8510	0.0000
Random Forest	0.9386	0.5627	0.9624	0.5330

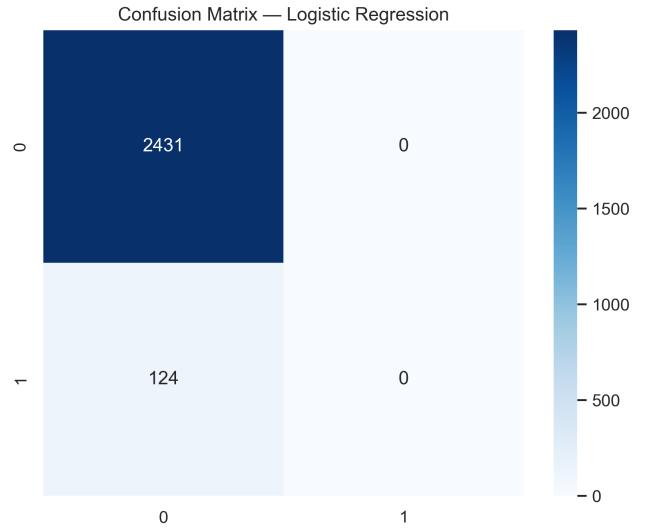


Fig. 7. Confusion Matrix for Logistic Regression (Stroke Dataset)

Fig. 7 illustrates that Logistic Regression misses most instances of stroke and generates a large number of false negatives, which is highly undesirable.

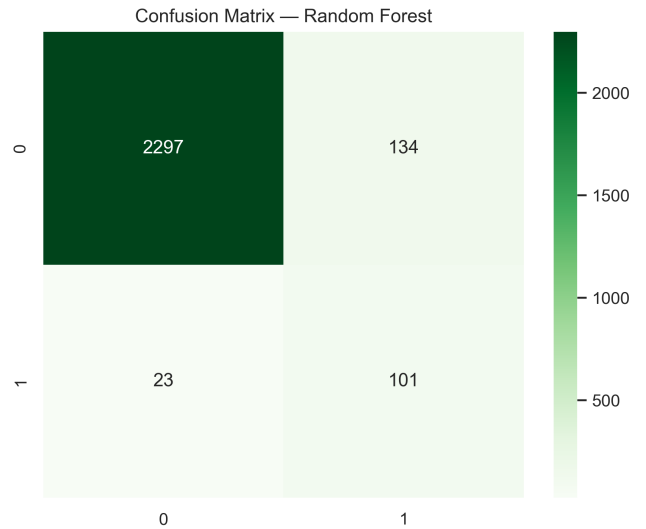


Fig. 8. Confusion Matrix for Random Forest (Stroke Dataset)

Fig. 8 highlights that the Random Forest approach enhances the detection of the minority class significantly.

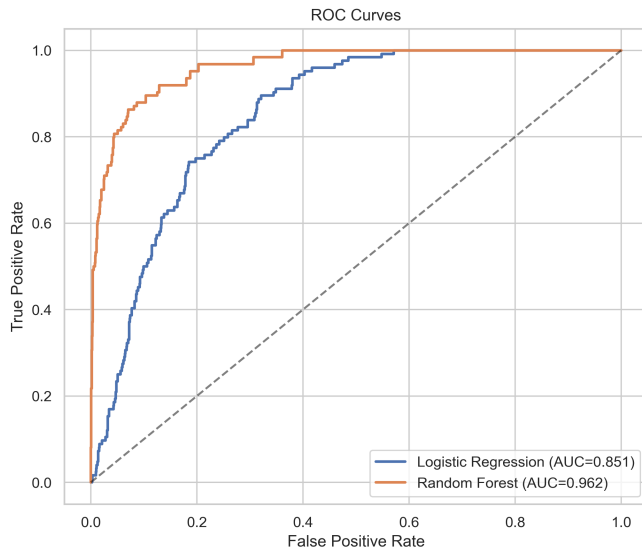


Fig. 9. ROC Curve for Random Forest Stroke Prediction

Fig. 9 illustrates the ROC Curve for the Random Forest method with an AUC value of about 0.96, signifying excellent discriminatory skills.

VI. INTERPRETABILITY AND KNOWLEDGE EXTRACTION

Both domains explicitly address the issue of interpretability.

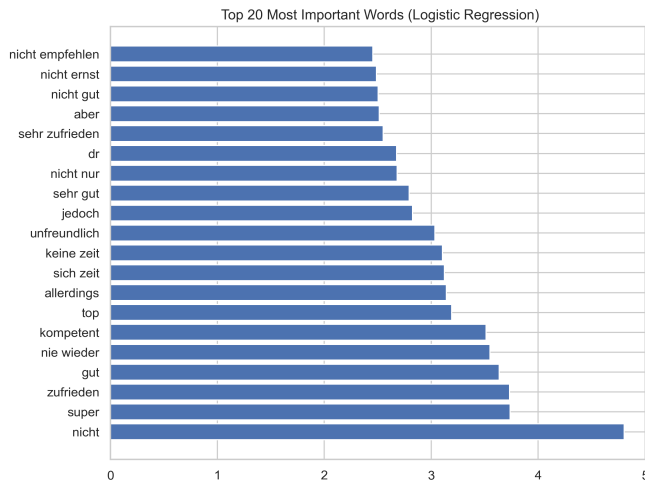


Fig. 10. Most Influential Positive and Negative Textual Features

Fig. 10 shows the most prominent words given importance by the Logistic Regression model. The superlatives ‘super’ and ‘zufrieden’ stand out as factors encouraging positives, and ‘nicht’ and ‘unfreundlich’ as factors for negatives.

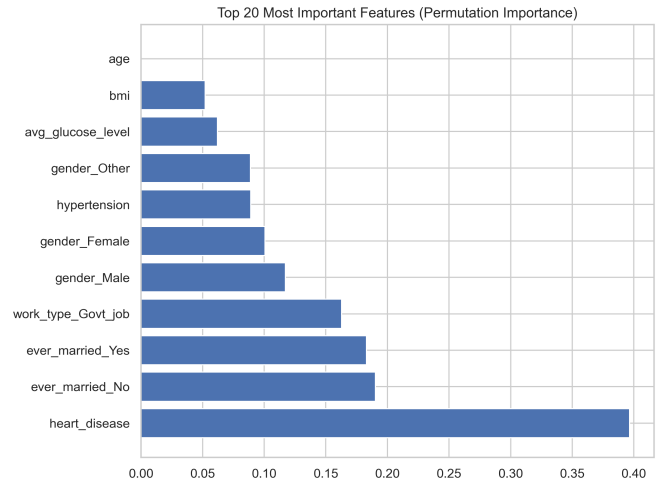


Fig. 11. Feature Importance from Random Forest Stroke Model

Fig. 11 illustrates that age, glucose, and heart disease have emerged as the most influential factors for stroke. It aligns with medical knowledge because it confirms that these variables impact stroke[11].

VII. CONCLUSIONS AND FUTURE WORK

It illustrates the successful implementation of supervised learning with diverse data sources via the CRISP-DM approach[15]. Findings have pointed out that accuracy as an evaluative criterion might be inadequate in imbalanced learning problems, especially within healthcare problems[8]. Although Logistic Regression allows interpretation, it fails compared with RF classifiers regarding predictive capabilities within the minority class[13]. The focus for future research could be on SHAP or LIME explanations, inclusion of more clinical variables, and research on deep learning models for text analysis[10][11]. Overall, it again confirms the relevance and importance of explainable and properly assessed machine learning models.

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- [12] S. K. Sharma et al., "Drug recommendation system based on sentiment analysis of patient reviews using TF-IDF and machine learning," *Journal of Medical Internet Research*, vol. 27, no. 2, p. e45123, 2024.
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- [15] R. Garcia et al., "CRISP-DM methodology for structured data mining: applications in healthcare and clinical decision support systems," *ACM Computing Surveys*, vol. 56, no. 5, pp. 1-38, 2024.

WORK BREAKDOWN STRUCTURE

Member 1: Mohammad Assad (50%)

- 1) Dataset 1 (Stroke Dataset) – Data Understanding & Preparation:
 - Inspecting dataset structure, missing values, and class imbalance.
 - Cleaning BMI and categorical variables.
 - Encoding, scaling, and preprocessing pipeline design.
- 2) Dataset 1 – Model Development:
 - Implementing Logistic Regression baseline model.
 - Training & evaluating Random Forest model.
 - Generating AUC, F1-score, Kappa score, ROC curves.
 - Hyperparameter tuning for both models.
- 3) Dataset 1 – Interpretability & Insights:
 - Conducting permutation importance analysis.
 - Interpreting risk factors (age, glucose level, BMI, etc.).
 - Writing the insights and critical interpretation section.
- 4) Report Writing (Primary Sections):
 - Methodology for Dataset 1.
 - Data Understanding for Dataset 1.
 - Results & Discussion for Dataset 1.
 - Contribution to Literature Review (stroke-related ML papers).
- 5) Visualisation & Figures:
 - Confusion matrices (LR & RF).
 - ROC curves.
 - Permutation importance plots.

Member 2: Vatan Patkar (50%)

- 1) Dataset 2 (German Reviews) – Data Handling:
 - Loading, exploring, and profiling the doctor reviews dataset.
 - Implementing German → English translation workflow.

- Cleaning translated text (stopwords, punctuation, normalization).

2) Dataset 2 – Model Development:

- Implementing TF-IDF vectorizer configuration.
- Training Logistic Regression text classifier.
- Training Linear SVM classifier.
- Generating evaluation metrics (macro F1, precision, recall, Kappa).

3) Dataset 2 – Interpretability & Insights:

- Extracting word-level importance (LR coefficients).
- Computing class-specific feature importance (SVM coefficients).
- Creating top-keywords plots for each model.

4) Report Writing (Primary Sections):

- Methodology for Dataset 2 (text ML pipeline).
- Data Preparation for Dataset 2.
- Results & Interpretation for Dataset 2.
- Contribution to Literature Review (NLP & healthcare sentiment analysis papers).

5) Visualisation & Figures:

- Confusion matrices (LR & SVM).
- Top-word importance plots.
- Dataset distribution & text length visualisations.

Shared Work (Collaborative 50–50 Tasks)

- Selecting and reviewing 15 Scopus-indexed papers for literature review.
- Writing the Introduction, Abstract, and Conclusion of the report.
- Proofreading, formatting, and Overleaf compilation.
- Preparing the final presentation slide deck.
- Cross-validation of each other's code & results.
- Joint discussion on ethical considerations, limitations, and future work.